AWS Setup and Security Alerting System

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Project Overview:

In this project, an AWS Free Tier account was opened, and security alerting mechanisms leveraging EventBridge, SNS (Simple Notification Service), and AWS Cloud Trail were implemented. The primary objective was to monitor and generate alerts upon detection of GetCallerIdentity API requests initiated by the AWS root user, which could indicate unauthorized access attempts or security vulnerabilities.

Objectives:

1. Creation of an AWS Free Tier account to conduct experiments in the cloud.

2. Make sure CloudTrail is set up to record AWS API activities.

3. SNS (Simple Notification Service) from Amazon can be used to deliver email notifications.

4. Configure Amazon EventBridge so that root user actions will cause alerts to be sent.

5. To verify functionality, use the AWS CLI to test the configuration.

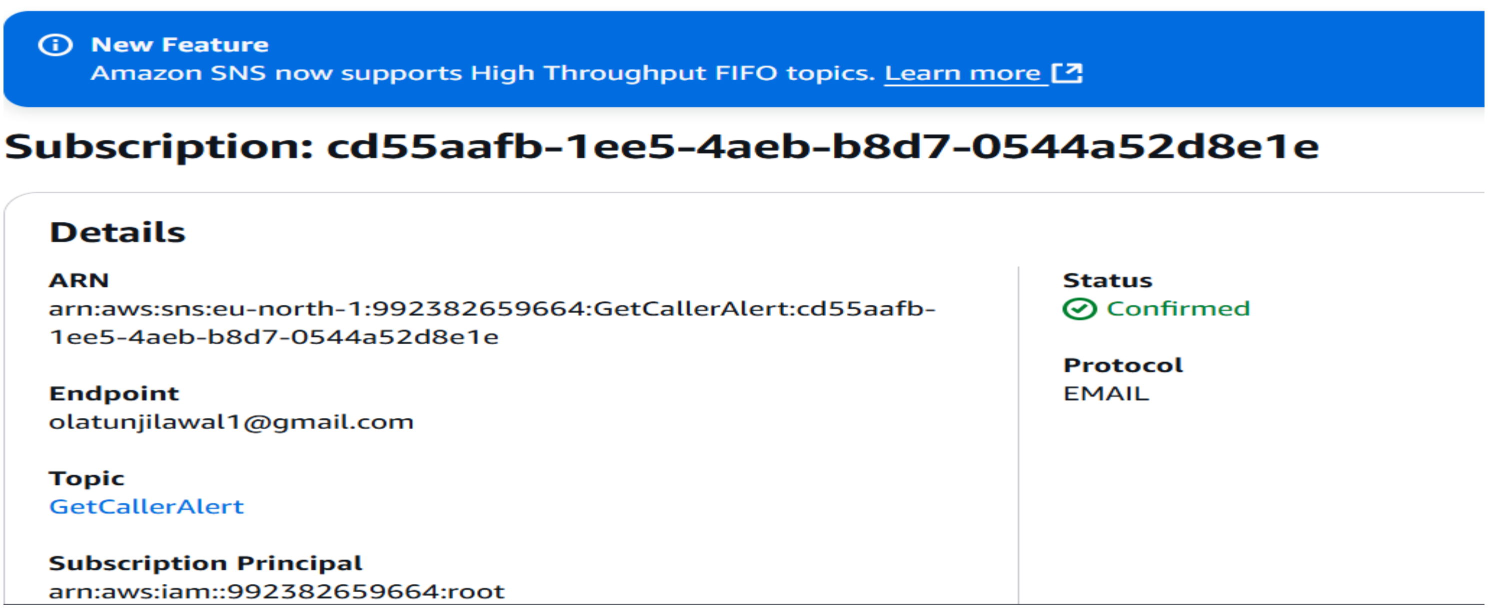
Technologies Used:

- **AWS CloudTrail**: Tracks API calls and logs events.
- **Amazon SNS**: Sends notifications to subscribed users.
- **Amazon EventBridge**: Creates event-driven triggers.
- **Kali Linux CLI**: Used to test the security setup.

Project Steps:

1. AWS Free Tier Account Setup

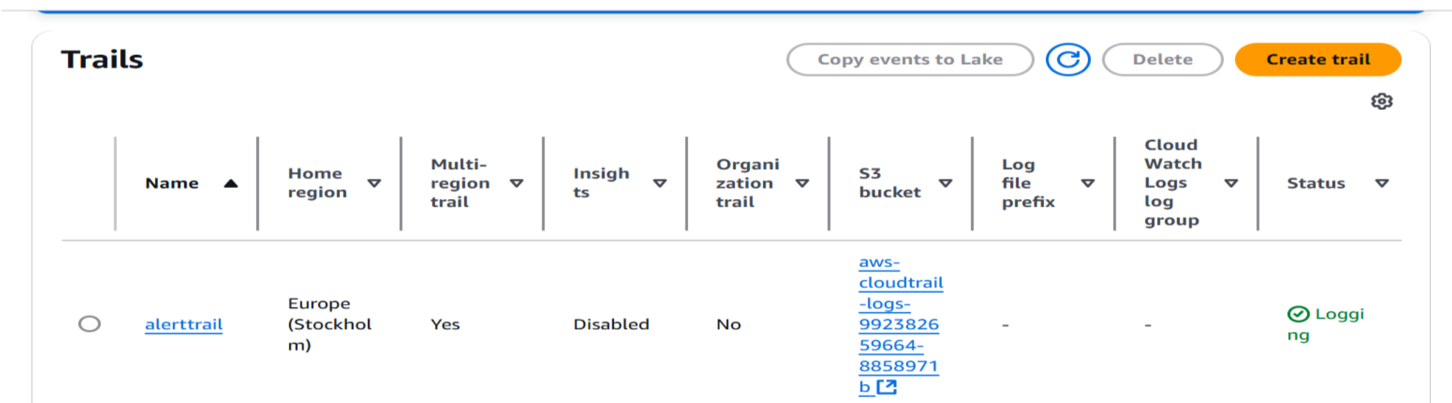
- Created a Free Tier AWS account on aws.amazon.com.
- Configured billing information and identity verification.
- Selected "Basic Support – Free" for cost efficiency.

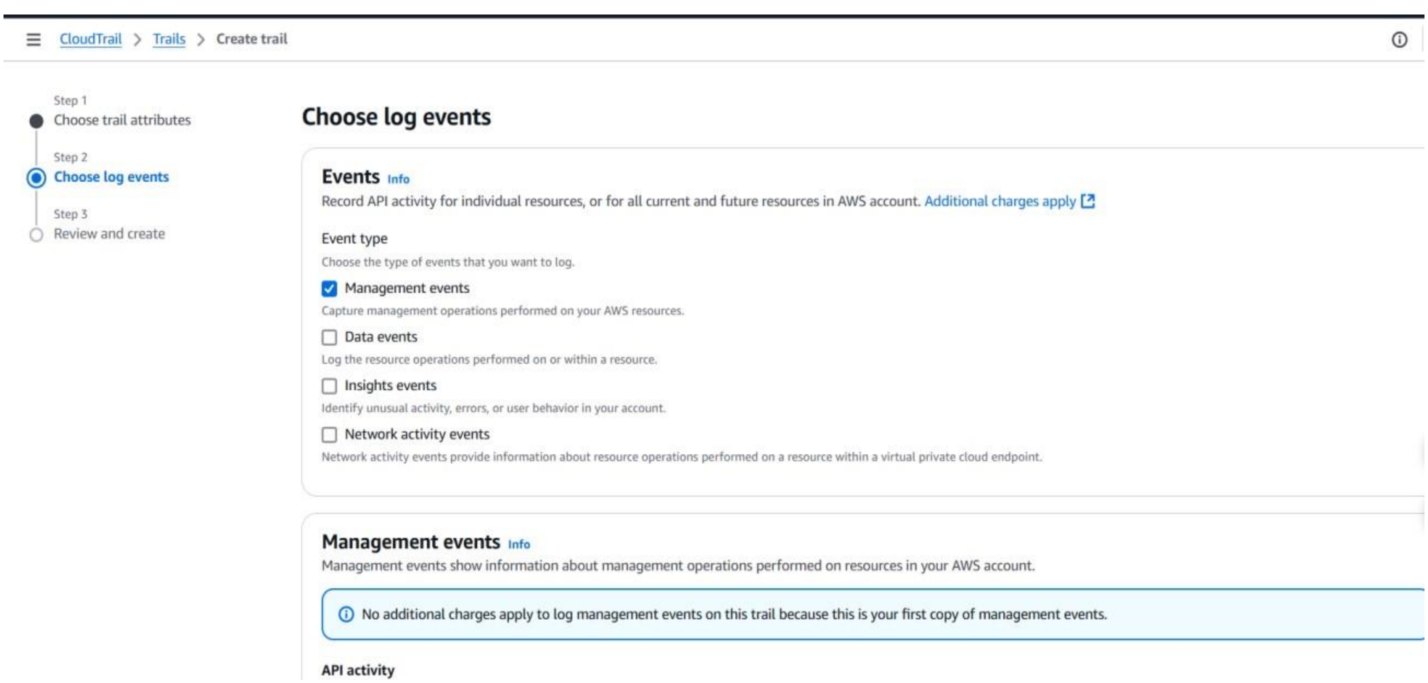


2. Configured AWS CloudTrail

- Accessed **CloudTrail** in the AWS Management Console.

- Created a new trail named MyCloudTrail.
- Enabled **Management Event Logging** for all read/write events.
- Configured logs to be stored in an **S3 bucket** (my-cloudtrail-logs-uniqueid).
- Enabled **multi-region logging** for full coverage.

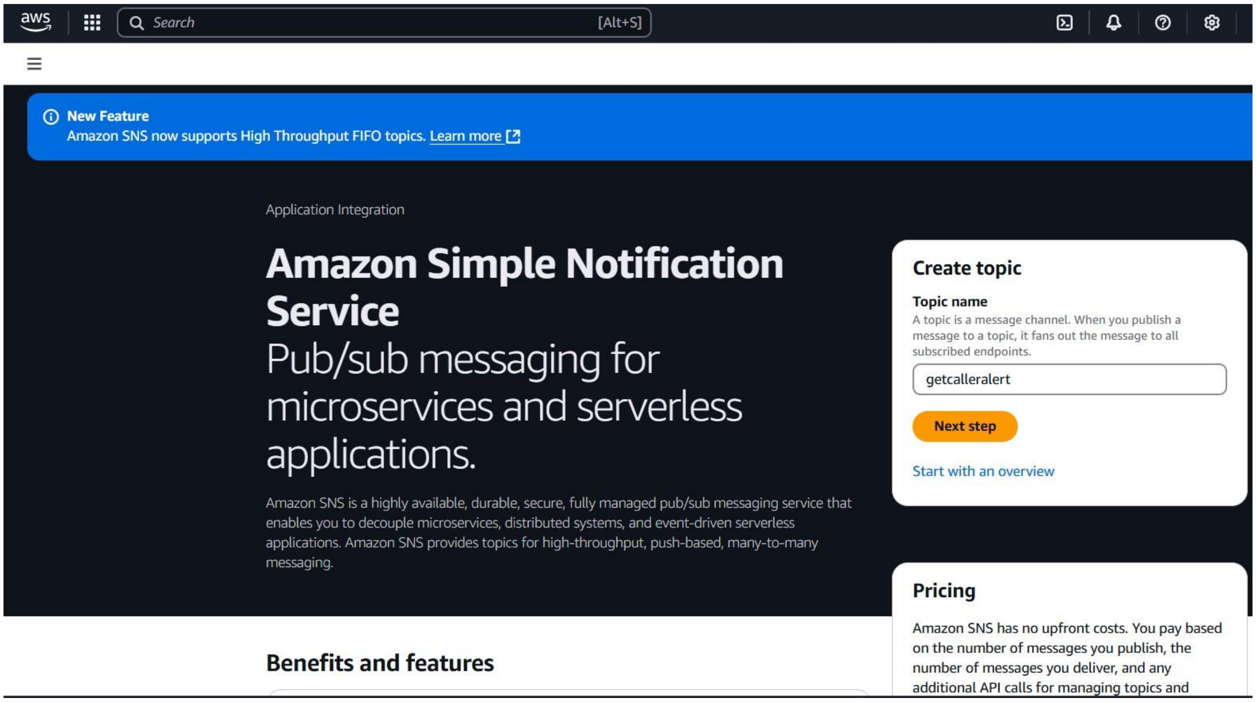




3. Set up SNS for Email Alerts

- Created an **SNS topic** named APICallAlert.
- Subscribed an email address to receive notifications.

- Confirmed the email subscription via the AWS verification link.
- Linked **CloudTrail** to SNS to trigger notifications when specific events occur.

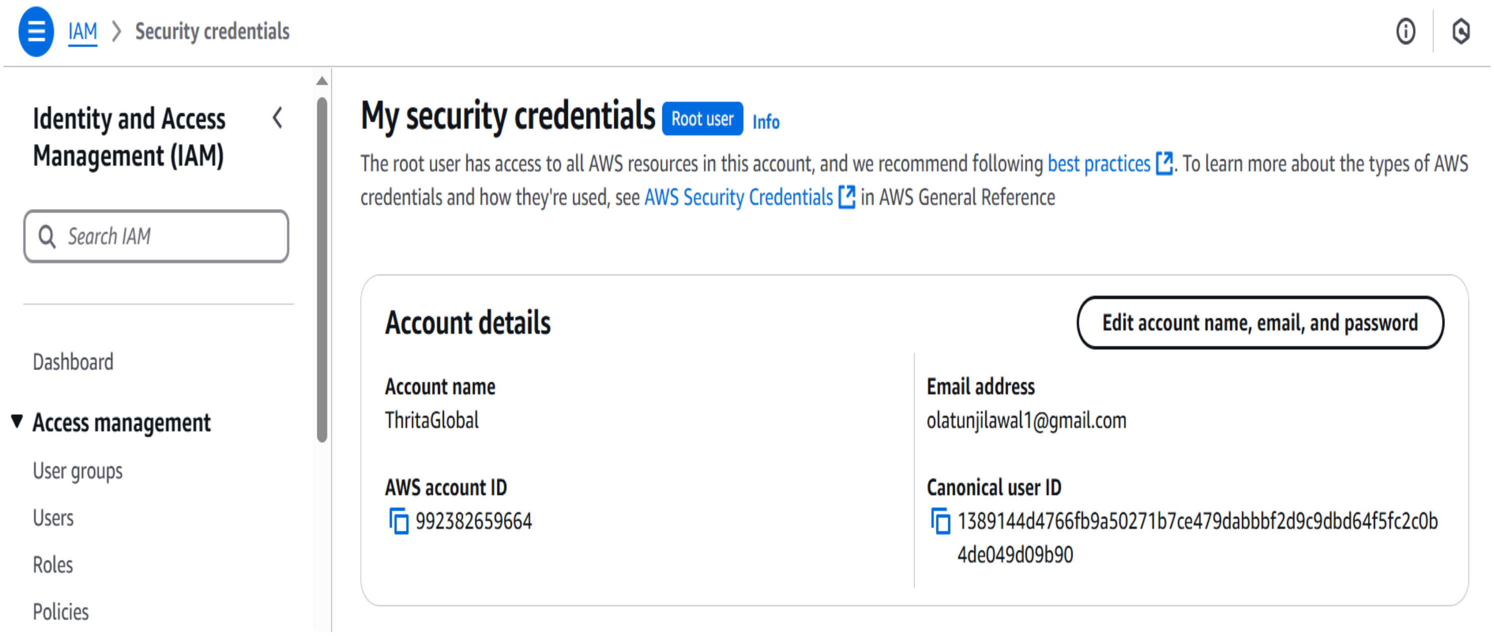


4. Configuring EventBridge for Root API Call Alerts

- Created an **EventBridge Rule** named GetCallerIdentityAlert.
- Defined a **custom event pattern** to detect GetCallerIdentity API calls from the root user:

```
{
 "source": ["aws.sts"],
 "detail-type": ["AWS API Call via CloudTrail"],
 "detail": {
  "eventSource": ["sts.amazonaws.com"],
  "eventName": ["GetCallerIdentity"],
  "userIdentity": {
   "type": ["Root"]
  }
 } }
```

- Configured **SNS** as the target for this rule to send notifications.
- Allowed EventBridge to create an IAM role for permissions.

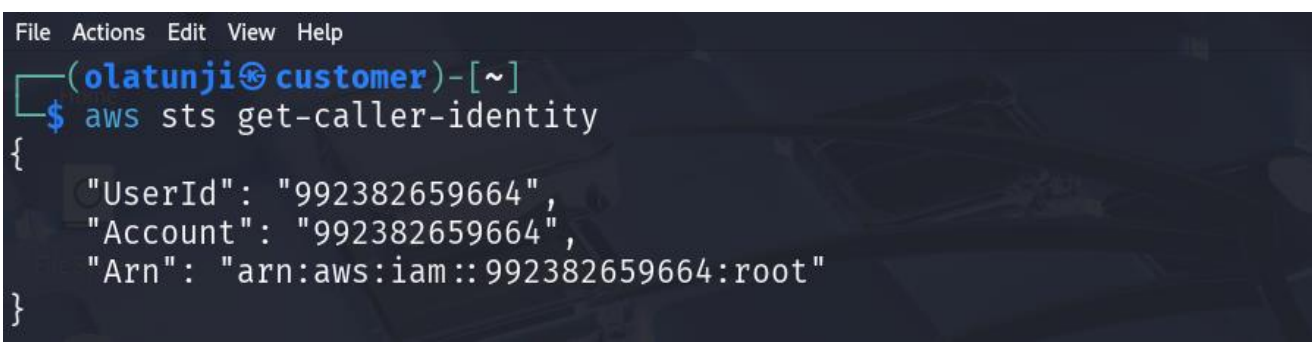


5. Testing the Setup via kali Linux CLI

- Installed and configured **kali linux** with root credentials.
- Ran the following command to simulate an API call:

  aws sts get-caller-identity

- Verified that the API call was logged in **CloudTrail**.
- Checked the SNS email inbox for the alert notification.



Key Takeaways

- Successfully implemented **real-time monitoring** for root user API calls.
- Demonstrated how to configure **AWS security alerting** using CloudTrail, SNS, and EventBridge.
- Gained hands-on experience with AWS **event-driven automation**.
- Ensured that **email notifications** were triggered on unauthorized root account activity.

This project demonstrates my capacity to successfully monitor cloud infrastructure and apply AWS security controls.